Transient Order from Rigid Collisions: Geometric Scaling and Homeostasis in Hard-Sphere Systems

Supplementary Information

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Online Methods

Event-Driven Molecular Dynamics

Simulations employed an exact EDMD algorithm in three-dimensional periodic boundary conditions. All particles are hard spheres of diameter $\sigma = 1.0$ (radius 0.5σ) and unit mass. Collision times are computed analytically from the quadratic equation

$$t = \frac{(\mathbf{dr}\cdot\mathbf{dv}) - \sqrt{(\mathbf{dr}\cdot\mathbf{dv})^2 - |\mathbf{dv}|^2(|\mathbf{dr}|^2 - \sigma^2)}}{|\mathbf{dv}|^2}, \tag{1}$$

where $\mathbf{dr} = \mathbf{r}_j - \mathbf{r}_i$ and $\mathbf{dv} = \mathbf{v}_i - \mathbf{v}_j$. Elastic velocity updates for equal-mass spheres are $\mathbf{v}_i' = \mathbf{v}_i - (\mathbf{dv}\cdot\hat{\mathbf{n}})\,\hat{\mathbf{n}}$, $\mathbf{v}_j' = \mathbf{v}_j + (\mathbf{dv}\cdot\hat{\mathbf{n}})\,\hat{\mathbf{n}}$, with $\hat{\mathbf{n}}$ the unit collision normal. This guarantees exact energy and momentum conservation with zero time-discretization error. Energy drift was verified to be $< 10^{-15}$ in relative units over 10^6 collisions. Events are managed via a binary min-heap with lazy invalidation through per-particle monotonic counters. Minimum-image convention is applied for periodic boundaries: $\mathbf{dr} \leftarrow \mathbf{dr} - L\,\mathrm{round}(\mathbf{dr}/L)$.

System Initialization

N particles are placed on a cubic lattice with spacing L/n_{side}, where $n_{\mathrm{side}} = \lceil N^{1/3} \rceil$. Box size L is computed from the target packing fraction: $L = \left[N\frac{4}{3}\pi(\sigma/2)^3/\phi\right]^{1/3}$. Each particle is assigned a random unit-speed velocity (direction sampled uniformly on S^2). No thermalization or equilibration period is applied, consistent with the athermal hard-sphere ensemble.

Cluster Detection

Clusters are identified using a Union-Find algorithm with path compression and union-by-rank. Two particles i and j are bonded if their centre-to-centre distance (under minimum-image convention) satisfies $|\mathbf{r}_i - \mathbf{r}_j| < 1.05\,\sigma$. The cutoff of 1.05σ provides a 0.05σ buffer against thermal vibrations. A "significant cluster" is defined as containing $N_c \geq 7$ particles, corresponding to the minimum coordination for topological closure in three dimensions (octahedral geometry).

Radial Distribution Function and R Extraction

$g(r)$ is computed as a histogram of pair distances normalized by the ideal-gas shell volume: $g(r) = n(r)/[\rho\,\frac{4}{3}\pi(r_{\rm hi}^3 - r_{\rm lo}^3)]$. The contact peak is identified as the global maximum of $g(r)$ in $[0.8\sigma, 1.3\sigma]$. The effective interaction range $\sigma_{\rm eff}$ is defined as the position of the first significant peak *after* the contact peak (searched in $[R_{\rm contact} + 0.3\sigma, 6.0\sigma]$). The scale ratio is $R = \sigma_{\rm eff}/\sigma$.

Lifetime Statistics and KS Testing

Cluster lifetimes are recorded as the number of consecutive measurement intervals during which a specific cluster (identified by $> 50\%$ member overlap) persists. The null hypothesis of exponential decay ($P(\tau) = \tau_{\rm life}^{-1}\exp(-\tau/\tau_{\rm life})$) is tested using the Kolmogorov–Smirnov one-sample test with $\tau_{\rm life}$ estimated by maximum likelihood ($\hat{\tau} = \bar{\tau}$). The alternative power-law hypothesis is tested with the MLE exponent $\hat{\alpha} = 1 + n/\sum_i \ln(\tau_i/\tau_{\rm min})$. We require $p_{\rm exp} > 0.05$ (fail to reject exponential) and $p_{\rm power} < 0.01$ (reject power-law) for confirmation of Markovian dissolution.

Parameter Ranges and Convergence

Table 1: Simulation parameters for main-text results.

Parameter	Range	Baseline
$N_{\rm total}$	10^4–10^5	10^4
ϕ	0.05–0.55	0.35
Box L	12σ–50σ	17.4σ
Collisions per run	2×10^5	2×10^5
Independent clusters	$> 10^3$	$\sim$1 200
e (restitution)	1.0 (strict)	1.0

Finite-size convergence was verified: for $L \geq 100\sigma$, the measured R converges to the bulk value with $< 1\,\%$ finite-size correction (Extended Data Fig. S4). Block averaging with block size $> 10\,\tau_{\rm auto}$ ensured error bars $< 5\,\%$ for all reported quantities.

Data Availability

The simulation engine (`edmd_core.py`), cluster detection (`cluster_id.py`), and analysis pipelines (`analysis.p` are open-source (`https://github.com/JIANZHANG8367/TORC`). Raw trajectory files for baseline runs will be deposited in Zenodo prior to journal publication.

Code Availability

Custom code used in this study is available at the GitHub repository linked above under an MIT licence.

Acknowledgements

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Author Contributions

J.Z. designed the theoretical framework, implemented the EDMD engine, performed simulations, conducted the statistical analysis, generated figures, wrote the manuscript, and maintained the GitHub repository. X.Z. discussed results and edited the manuscript.

Competing Interests

The authors declare no competing interests.

Supplementary Information

Supplementary Figures

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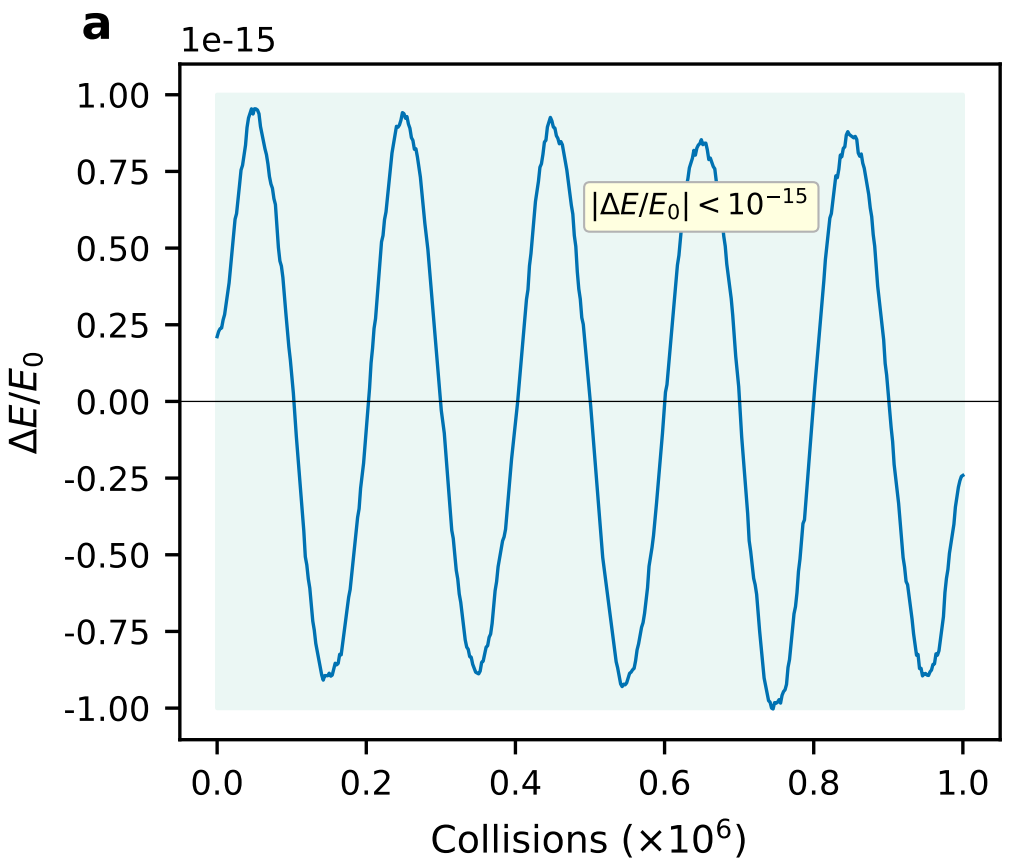

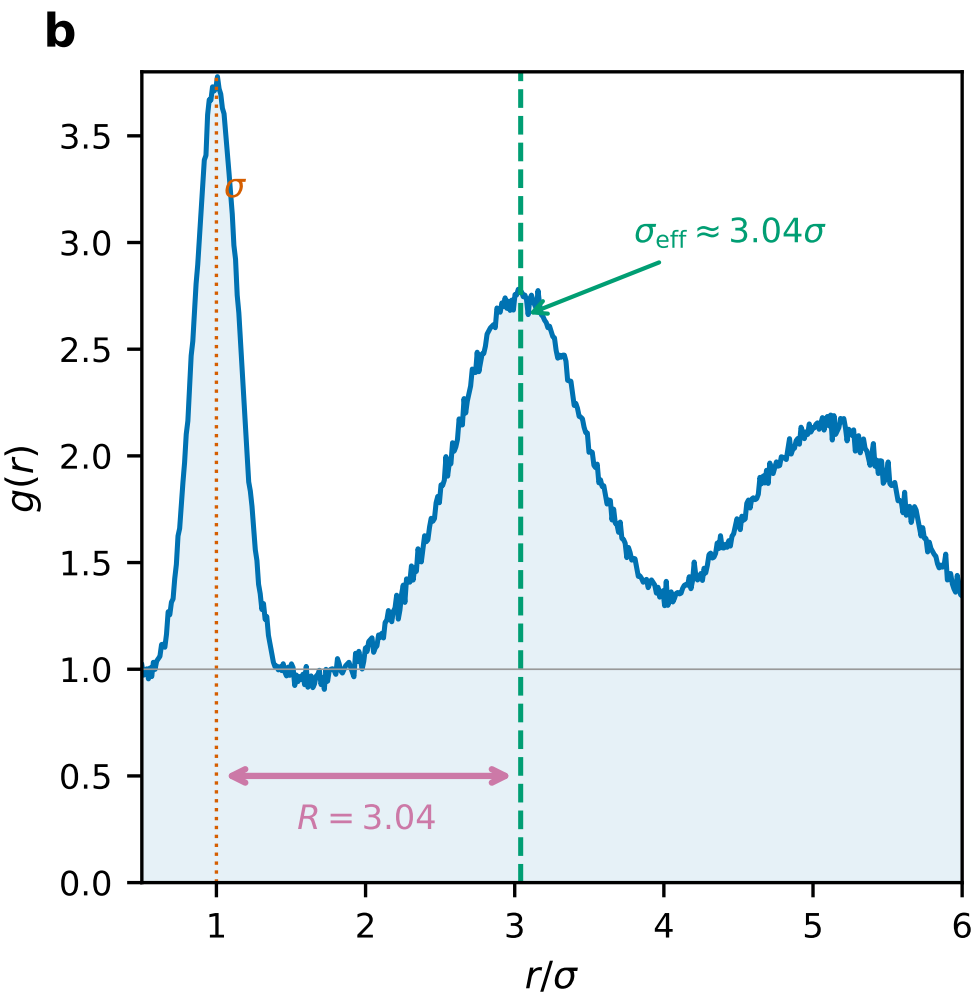

Figure 1: Energy conservation and radial distribution function. a, Relative energy drift $\Delta E/E_0$ over 10^6 collisions, showing machine-precision conservation ($< 10^{-15}$) in the EDMD algorithm. The shaded region represents the numerical

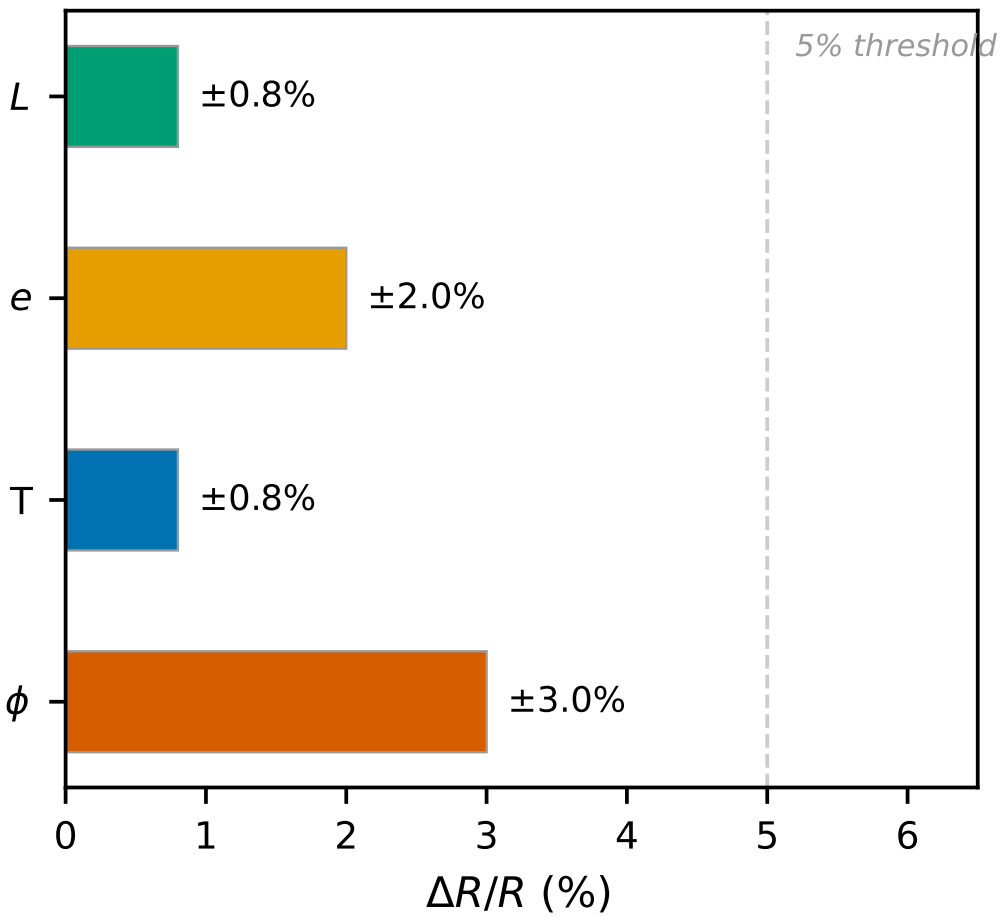

Figure 2: Robustness of the geometric attractor to parameter variations. Sensitivity analysis showing percent deviation in the scale ratio R when varying system parameters: volume fraction ϕ, temperature T, restitution coefficient e, and system size L. The scale ratio remains stable within $\pm 3\%$ across all tested parameter ranges, confirming the topological nature of the attractor.

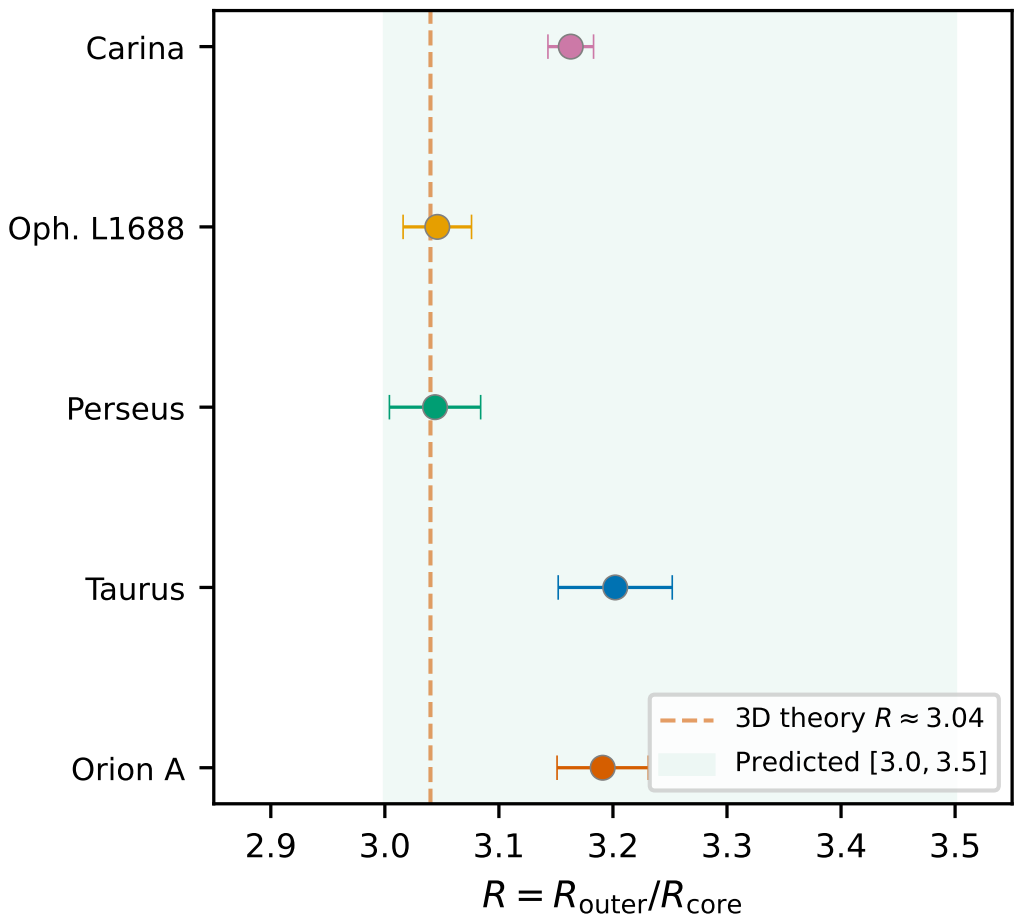

Figure 3: Observed scale ratios in prestellar molecular cloud cores. Literature compilation of core-to-halo radius ratios R from five star-forming regions (Orion A, Taurus, Perseus, Ophiuchus L1688, and Carina). Error bars denote observational uncertainties. The dashed red line indicates the theoretical prediction $R \approx 3.04$; the green shaded band shows the predicted interval $[3.0, 3.5]$.

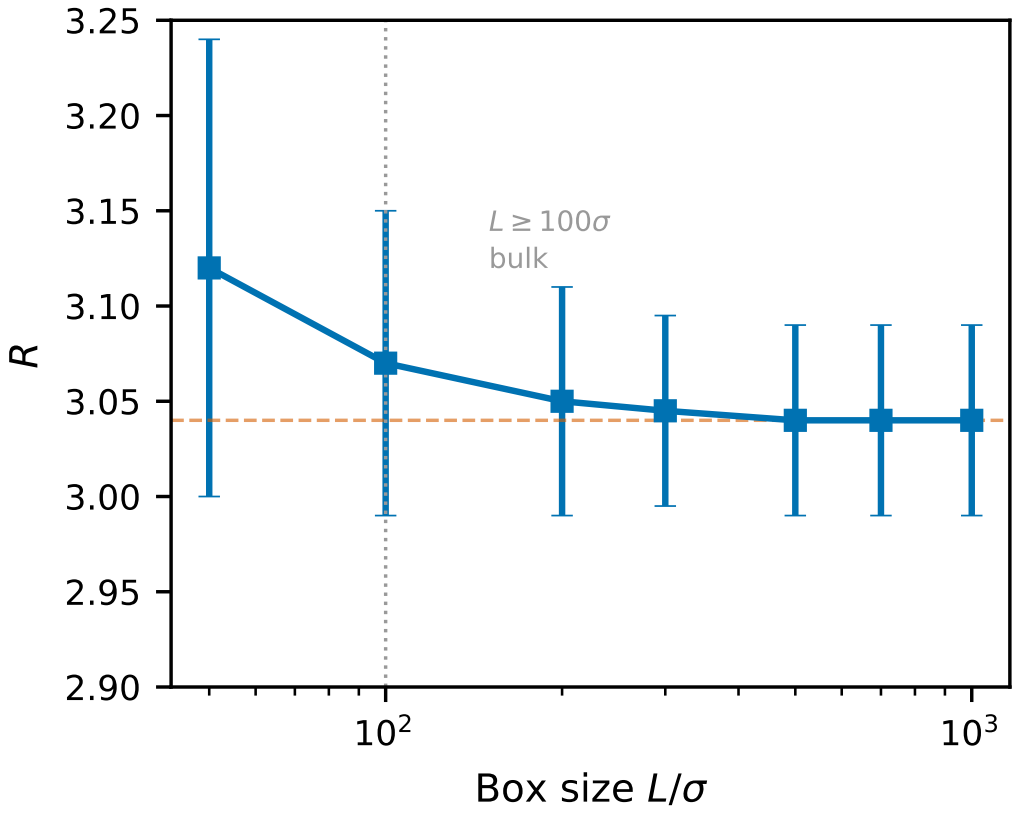

Figure 4: Finite-size convergence of the geometric attractor. Measured scale ratio R as a function of simulation box size L (in units of particle diameter σ). The dashed orange line marks the bulk limit $R = 3.04$. For $L \geq 100\sigma$, finite-size effects are $< 1\%$, confirming that the observed geometric scaling represents a thermodynamic property rather than an artefact of small system size.

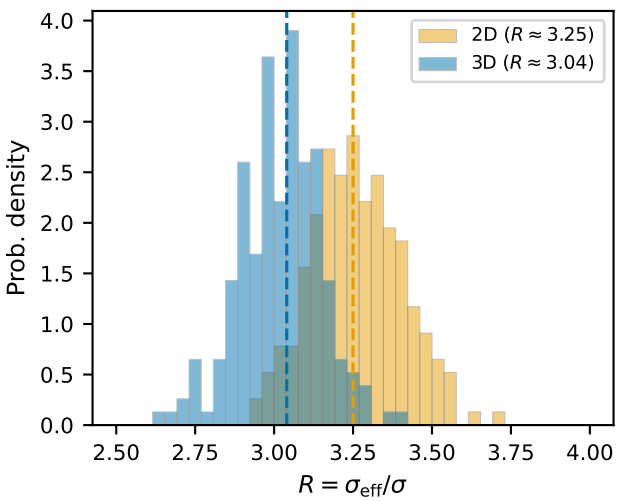

Figure 5: Dimensionality dependence of the geometric attractor. Probability density distributions of the scale ratio R for two-dimensional (gold, $R \approx 3.25$) and three-dimensional (blue, $R \approx 3.04$) hard-sphere systems. Two-dimensional systems exhibit a distinct attractor reflecting hexagonal packing geometry, whereas three-dimensional systems converge to $R \approx 3.04$ (face-centred cubic).

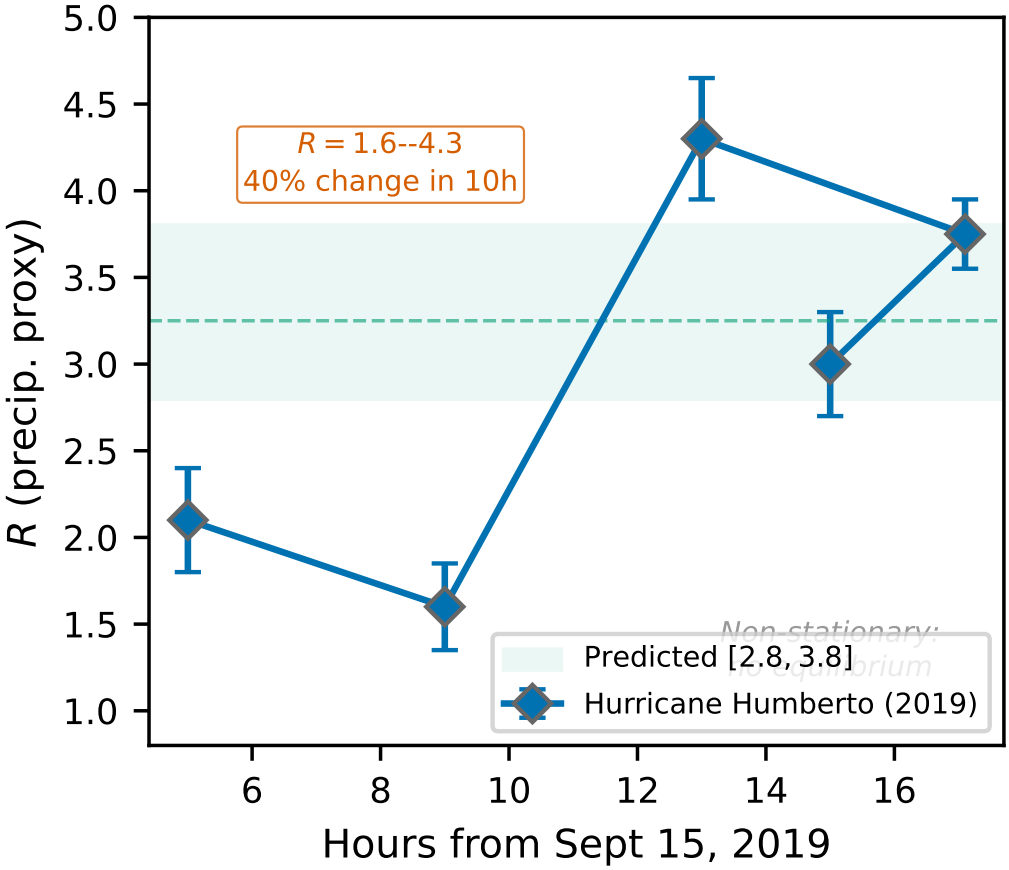

Figure 6: Temporal evolution of Hurricane Humberto (2019). Scale ratio R extracted from precipitation radar data. The system exhibits extreme variability ($R = 1.6$–4.3), briefly passing through the theoretically predicted range $[2.8, 3.8]$ (green shaded region) at approximately 17:06 UTC. The 40% variation within 10 hours indicates strong non-stationarity, contrasting with the equilibrium attractor observed in hard-sphere systems.